

Task Division — Talk-to-my-Data

Rotation across all 4 pillars: everyone takes a different sub-task each pillar, so by Stage 3 everyone has touched data, AI, backend, and frontend. Tasks marked Required come directly from Artefact's brief — don't skip these. Tasks marked Stretch are what makes the project stand out, but are the first things to cut if the timeline gets tight.

Stage 0 — Prerequisites

Sub-task	Owner	Priority	Deadline
0.1 Cloud Setup (GCP project, CLI, service decisions)	Raed	Required	Sat, Sep 12
0.2 Environment Setup (Python, venv, IDE, .gitignore)	Muhammed	Required	Sat, Sep 12
0.3 Architecture Sketch v1 (Excalidraw)	Amir	Required	Sat, Sep 12
0.4 Dataset & KPI Draft	Fares	Required	Sat, Sep 12

Gate: GCP project exists, environment documented, first sketch drawn, dataset downloaded.

Stage 1 — Contracts & Skeleton ☒ COMPLETE

Sub-task	Owner	Priority	Deadline
1.1 Write schema.md	All (together)	✔ Done	Closed
1.2 Write api.md	All (together)	✔ Done	Closed
1.3 Repo Skeleton (/src, /mock_data, /docs)	All (together)	✔ Done	Closed

Gate cleared: both contract files committed and rendering correctly, folder skeleton matches the plan.

Stage 2 — Core Build

Pillar 1 — Cloud & Data Platform

Sub-task	Owner	Priority	Deadline
1.1 Retail Data Cleaning & ETL	Raed	Required	Sat, Sep 19
1.2 BigQuery Data Warehouse Setup	Fares	Required	Sat, Sep 19
1.3 IAM & Security Configuration	Amir	Stretch	Sat, Sep 19
1.4 Docker & Cloud Run Deployment	Muhammed	Required (minimal for now)	Sat, Sep 19

Gate: real data queryable in BigQuery by everyone.

Pillar 2 — LLM & Text-to-SQL

Sub-task	Owner	Priority	Deadline
2.1 Dynamic Schema Injection	Fares	Required — core ask	Sat, Sep 26
2.2 Few-Shot Prompting & Domain Lexicon	Amir	Stretch	Sat, Sep 26
2.3 Self-Healing Query Feedback Loop	Muhammed	Stretch	Sat, Sep 26
2.4 Ambiguity Resolution & Clarification	Raed	Stretch	Sat, Sep 26

Gate: a plain-English question reliably produces correct SQL against real data.

Pillar 3 — Backend & Analytics Engine

Sub-task	Owner	Priority	Deadline
3.1 FastAPI Core Service Layer	Amir	Required	Sat, Sep 26
3.2 SQL Validation & Safety Guardrails	Muhammed	Required	Sat, Sep 26
3.3 Retail KPI Computation Engine	Raed	Required — brief asks for 2+ KPIs	Sat, Sep 26
3.4 Multi-Modal Response Formatter	Fares	Required	Sat, Sep 26

Gate: /ask endpoint works end-to-end, callable directly, no UI needed yet.

Pillar 4 — Frontend UI, Benchmark & Delivery

Sub-task	Owner	Priority	Deadline
4.1 Streamlit Conversational Interface	Muhammed	Required	Sat, Oct 3
4.2 Dynamic Data Visualizations	Raed	Stretch	Sat, Oct 3
4.3 Golden Benchmark Evaluation Suite	Fares	Stretch (10-15 questions is fine)	Sat, Oct 3
4.4 Excalidraw Architecture & Client Deck	Amir	Required — named in the brief	Sat, Oct 3

Gate: a real question typed into the real UI returns a real, working answer.

Stage 3 — Integration

Sub-task	Owner	Priority	Deadline
Full Pipeline Integration (replace remaining mocks, test end-to-end)	All (together)	Required	Wed, Oct 7

Gate: the full system works start to finish, without anyone manually patching things together.

Stage 4 — Dry Run Prep

Sub-task	Owner	Priority	Deadline
Dry Run Deck + Rehearsal, presented to Mark + Quinten	All (together)	Required	Wed, Oct 7

Gate: feedback received, clear list of what to fix before the final.

Stage 5 — Final Polish & Deployment

Sub-task	Owner	Priority	Deadline
Full Cloud Run Deployment + Final Deck + Live Demo Rehearsal	All (together)	Required	Wed, Oct 7

Gate: live demo works, deck is ready, everyone knows their speaking part.